A principle of balancing non-alternation and alternation in tonal metres

Abstract: Poetic metres govern the basic rhythmic patterns of verses and lines, and generally require the phonological property they regulate (e.g. stress, weight and/or tone) to alternate within a line to avoid monotony. Using the tonal metre of Middle Chinese regulated verse as an example, this paper proposes that a metre can also require the regulated property not to alternate, and that this requirement is balanced against alternation. The canonical templates of regulated verse require tone categories to alternate within a line, but only to a restricted extent: no non-final syllable in the templates is flanked by two syllables of the opposite category. Beyond the templates, the paper shows in Optimality Theory that the remedies prescribed by traditional prosody for irregular lines follow from the interaction of constraints encoding alternation and non-alternation. The two remedies for *guping* 'Lone Even' (孤平) are outputs of different rankings within a single, partially ranked grammar: the remedy within the line satisfies a non-alternation constraint against a lone Even tone, whereas the remedy in the other line of the couplet satisfies an alternation constraint requiring the two lines of a couplet to be tonal opposites (对). The 'three-inversion-four-remedy' pattern (三拗四救) arises when alternation in the cadence of a line takes priority over faithfulness. Finally, I show that this balancing property enriches the typology of poetic metres and parallels restrictions found in the tonal phonology of natural languages, and that metres can be informed by – and thus serve as a window into – natural language phonology.

Keywords: metrics; tonal languages; *guping*; Chinese poetry; regulated verse

1. Introduction

Poetic metres regulate the format of poetry in various respects, including prosody, rhyme and parallelism. From the prosodic perspective, metres govern the basic rhythmic patterns of verses and/or of the lines in verses. The metres of Indo-European languages such as English, French, Greek and Latin have been intensively studied by generative phonologists (e.g. Fabb & Halle, 2008; Kiparsky, 2020). Some metres even serve as a primary source of evidence for the phonology of a language, especially when the language is no longer spoken (e.g. Ancient Greek or Vedic Sanskrit). In contrast, metres for poetry in tonal languages remain relatively unexplored in the generative framework. Poems written in Chinese at most stages of its history also employ complex metres. Although these metres have been described extensively in traditional Chinese literature and analysed in a number of studies within the generative framework (e.g. Chen, 1979; Graham, 1980; Lorentz, 1980; Yip, 1980; Napoli, 1989, 1991; Xue, 1989; Duanmu, 2004; Duanmu & Stiennon, 2005), they have rarely been compared systematically with the tonal phonology of Chinese and other tonal languages.

In this paper, I analyse the metre of 'regulated verse' in Middle Chinese and propose that it calls for a new principle in generative metrics, namely the coexistence of non-alternation and alternation in the property that the metre regulates. The metrical templates of regulated verse dictate that tone categories must alternate within a line, but not too abruptly. I further show that the 'remedies' (拗救 *aojiu*) prescribed by traditional prosody for lines that deviate from the templates reveal how the two requirements are balanced against each other, and I argue that this feature of the metre mirrors general tonal preferences in natural languages.

The paper is arranged as follows. Section 2 introduces regulated verse, its metrical templates and previous analyses of its metrical structure. Section 3 presents the observations showing that the metre requires both alternation and non-alternation of tone categories. Section 4 analyses, in Optimality Theory, the traditional prohibition known as *guping* (孤平, 'Lone Even'; Duanmu & Stiennon, 2005: 5) together with its two remedies, the related remedy known as 'three-inversion-four-remedy' (三拗四救), and the prohibition of three Even tones at the end of a line (三平调). Section 5 discusses the implications of the analysis for metrical typology and for natural language phonology, the explanatory status of the analysis, and the status of *guping* as a convention or an intuition. Section 6 concludes.

2. Background[①]

2.1 Regulated verse and its tone categories

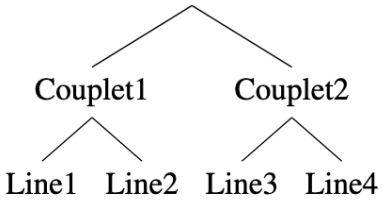
Regulated verse (近体诗) is a type of traditional Chinese poetry that took shape in the early Tang Dynasty and flourished throughout the Tang Dynasty (618–907 CE). Poems generally have four or eight lines: four-line poems are called *jueju* (绝句) and eight-line poems *lüshi* (律诗).[②] Each line has five or seven characters, i.e. syllables. Every two lines make up a couplet (联), and every four lines make up a quatrain (1).

[①] Unless otherwise stated, the descriptions of regulated verse follow 王力 (1958, 1962) and 冯胜利 (2015).

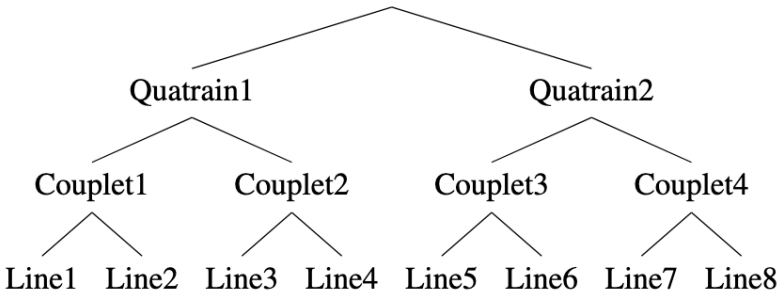
[②] Longer poems, called *pailü* (排律), also exist.

(1) How lines are grouped into couplets and quatrains in regulated verse

a. Four lines:



b. Eight lines:



Middle Chinese was spoken at the time when regulated verse became popular. There were four tones in Middle Chinese, which were divided into two categories for metrical purposes. The level (*píng* 平) tone constitutes its own category, 'Even' (*píngshēng* 平声), while the rising (*shǎng* 上), departing (*qù* 去, i.e. falling) and entering (*rù* 入, i.e. checked) tones make up the category 'Oblique' (*zèshēng* 仄声). I will represent the two categories as E and O respectively in what follows.

2.2 Canonical templates and free positions

Poets were required to follow rigid metrical templates that restrict the tone category allowed at each position of a line. Tone categories alternate not only within a line but also between the two lines of a couplet. Within a couplet, the even-numbered positions of the two lines carry opposite tone categories (对 *duì*); across couplets, the second line of one couplet and the first line of the next carry the same category in their even-numbered positions (粘 *nián*). The template of an eight-line poem is therefore obtained by doubling the template of the corresponding four-line poem with the same line length and the same starting tone, so this paper looks at four-line templates only.

The canonical templates are given in (2). They are classified by the number of syllables in a line (5 or 7) and by the tone category of the second syllable of the first line, which traditional prosody uses to distinguish 'Even-starting' (平起) from 'Oblique-starting' (仄起) poems; in the canonical templates, the first syllable of each line belongs to the same category as the second.[①] There are thus four templates for four-line poems, each consisting of two

[①]The templates in (2) are those in which the first line does not rhyme. When the first line rhymes (首句入韵, the norm in heptasyllabic poems), it takes the form of a rhyming line instead (type B instead of type C, or type D

couplet types. I label the couplet types Couplet-5E, Couplet-5O, Couplet-7E and Couplet-7O according to the line length and the category of the line-initial disyllable. The templates use four line types, which I label A–D according to their pentasyllabic forms: A = OOEEO (仄仄平平仄), B = EEOOE (平平仄仄平), C = EEEOO (平平平仄仄) and D = OOOEE (仄仄仄平平). Lines of types B and D end in E and are the rhyming lines; lines of types A and C end in O and do not rhyme. Couplet-5O consists of lines of types A and B, and Couplet-5E of lines of types C and D.

Heptasyllabic lines are extensions of pentasyllabic lines formed by adding, at the beginning, two syllables of the category opposite to that of the first two syllables of the pentasyllabic line. For example, the first line of Couplet-7E, EEOOEEO, is constructed by adding two E syllables at the beginning of OOEEO, the first line of Couplet-5O. Hence, I will focus primarily on the last five syllables of each line, and I will use the labels A–D for heptasyllabic lines as well (A = EEOOEEO, B = OOEEOOE, C = OOEEEOO, D = EEOOOEE).

(2) Canonical templates for regulated verse (free positions underlined; line types in parentheses)

	Even-starting (平起)		Oblique-starting (仄起)	
Pentasyllabic	a. Couplet-5E:	E̲EEOO (C) O̲OOEE (D)	b. Couplet-5O:	O̲OEEO (A) EEOOE (B)
	Couplet-5O:	O̲OEEO (A) EEOOE (B)	Couplet-5E:	E̲EEOO (C) O̲OOEE (D)
Heptasyllabic	c. Couplet-7E:	EE̲O̲OEEO (A) O̲OEEOOE (B)	d. Couplet-7O:	O̲OE̲EEOO (C) EE̲O̲OOEE (D)
	Couplet-7O:	O̲OE̲EEOO (C) EE̲O̲OOEE (D)	Couplet-7E:	EE̲O̲OEEO (A) O̲OEEOOE (B)

Like many other metres, the templates for regulated verse allow some freedom. Positions that allow the use of the tone category other than the canonical one (free positions) are underlined in (2).[①] The last position of each line is never free, which is also statistically confirmed by Duanmu & Stiennon (2005: 24–25). Analogously to the tradition of Western metrics, I call the use of the non-canonical tone category at a position an 'inversion'.

instead of type A; see below for the line types). Since no new line types are involved, the analysis applies equally to such poems.

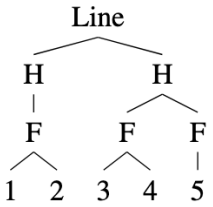
[①]There has long been a popular misconception that the free positions are the odd-numbered non-final positions of every line (positions 1 and 3 of pentasyllabic lines, and 1, 3 and 5 of heptasyllabic lines), as summarised by the saying 'Licence for 1, 3 and 5, strictness for 2, 4 and 6' (一三五不论，二四六分明). In fact, only the first position of pentasyllabic lines and the first and third positions of heptasyllabic lines consistently allow inversions (王力, 1962); the exceptions are discussed in Section 4.

2.3 Previous analyses of metrical structure

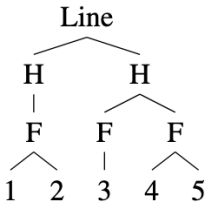
The templates for regulated verse have been analysed with linear (王力, 1958, 1962; Downer & Graham, 1963; T'sou, 1968; Jakobson, 1970; Lorentz, 1980; Napoli, 1991), hierarchical (Chen, 1979; Graham, 1980; Yip, 1980; Xue, 1989) and grid-based (Napoli, 1989) approaches. Although these analyses differ significantly in their treatment of a line, most studies since Chen (1979) agree that every line can be divided into two half-lines or hemistiches (H), and every hemistich into feet (F). In Chen's (1979) analysis, the two syllables of each foot carry tones from the same category, so that pentasyllabic and heptasyllabic lines receive the structures in (3), in which the position of the monosyllabic foot depends on the tone pattern of the line.

(3) Subline structure by Chen (1979)

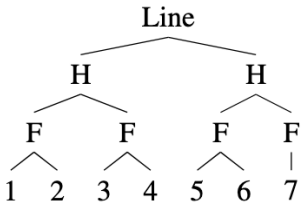
a. OOEEO and EEOOE lines:



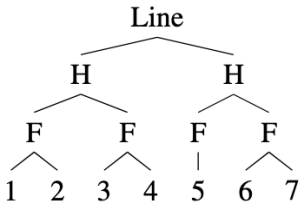
b. OOOEE and EEEOO lines:



c. OOEEOOE and EEOOEEO lines:



d. OOEEEOO and EEOOOEE lines:



(adapted from Chen, 1979:380)

This tone-based parsing has been criticised by Duanmu (2004: 71–72) (see also 端木三, 2016), who points out that there is no independent evidence for the trisyllabic or tone-dependent groupings, and that Chen's analysis allows monosyllabic feet and stress clash. Duanmu instead assigns all lines a uniform rhythmic template, in which disyllabic feet are followed by the final syllable and an empty beat (see (4) in Section 3.1). In the present analysis, I only adopt Chen's (1979) half-line division and reject the foot structure. The generalisations and constraints proposed in Sections 3 and 4 refer only to (i) the linear adjacency of tone categories, (ii) the final three syllables of a line, which I call the 'cadence' and which correspond to the traditional division of lines into an initial part and a final three syllables (上二下三 for pentasyllabic and 上四下三 for heptasyllabic lines), and (iii) the parity of positions (odd vs. even).

3. Observations

Given the basics of the metrical templates, this section shows that the tone categories of the syllables in each line need to alternate and not to alternate at the same time.

3.1 Alternation

The feature that most straightforwardly illustrates the need for alternation in regulated verse is the cap on the number of consecutive syllables from the same tone category. Because the syllables of the initial disyllable(s) of every line belong to the same category, it is common to find several syllables of the same category in a row. However, the number of consecutive syllables from the same tone category in the canonical templates never exceeds three. Moreover, runs of three occur only at the beginning of the five-syllable core of a line: in Couplet-5E, the first line begins with three E syllables (EEEOO) and the second line with three O syllables (OOOEE), and the same holds for positions 3–5 of the two lines of Couplet-7O. No canonical line ends in three syllables of the same category: the cadence always contains both categories (A: EEO, B: OOE, C: EOO, D: OEE).

Traditional prosody confirms that this is not accidental. A line ending in three Even tones (三平调 *sanpingdiao* 'three-Even ending'), which would arise from an inversion at position 3 of a type-D line (OOEEE), is strictly avoided in regulated verse (Section 4.6), although it is common in Old Style verse (古体诗; 王力, 1958: §28, 1962: §6). A line ending in three Oblique tones (三仄尾 *sanzewei* 'three-Oblique ending'), which could arise from an inversion at position 3 of a type-C line (EEOOO), is avoided less strictly. Thus, three is a maximum rather than a uniform allowance: it is reached only at the beginning of the five-syllable core, and the requirement of variation is strongest at the end of the line. In rhyming lines, the cadence always contains both categories; in non-rhyming lines, a uniform Oblique cadence is tolerated but dispreferred. Irregular lines, such as the five-Oblique line 向晚意不适 (仄仄仄仄仄) in 《登乐游原》by 李商隐, can exceed this maximum.

The other aspect that demonstrates the necessity of alternation is more subtle and subject to different analyses. Like accentual and quantitative metres, the metrical templates of regulated verse are believed to consist of positions that alternate in prominence. However, unlike in metres that are based on a natural phonological asymmetry such as stressed vs. unstressed or heavy vs. light syllables, in Chinese verse it is difficult to tell which position in a pair of adjacent positions is stronger.

Phonologists have thus diverged in their analyses of relative prominence. Chen (1979: 396) and Yip (1980: 108) both argued for iambic feet, partly on the basis of the recitation rhythm of the verse and partly on the assumption that the more prominent positions are those with fewer tonal choices. Napoli (1991: 254–257) argued that the recitation rhythm cannot directly explain the tonal patterns of the templates, because the chanting rhythm is independent of tones in Chinese verse in general (e.g. Boyce, 1980). Duanmu (2004; see also 端木三, 2016)

rejected the iambic analysis on both counts: recitation should not be used as evidence, and stressed syllables in Chinese have more, rather than fewer, tonal choices. He therefore proposed a trochaic template in which the odd-numbered positions are strong (4).

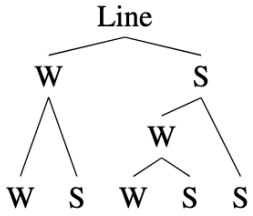
(4) Trochaic template (Duanmu, 2004: 55; S = strong, W = weak; Ø = empty beat)

a. Pentasyllabic lines: S W S W S Ø
b. Heptasyllabic lines: S W S W S W S Ø

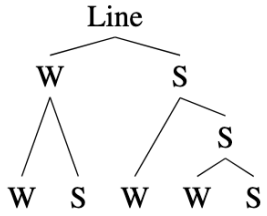
Despite these disagreements, analysts agree that there is a correlation between free positions (i.e. those that allow inversions) and the relative prominence of positions, even though they disagree on its direction. In the arboreal analyses of Chen (1979) and Yip (1980), the positions that allow inversion are the weak, odd-numbered positions of iambic feet (5). In the grid analysis of Napoli (1989), the free positions tend to be those that receive a mark on the second level of the grid (6), which are the more prominent positions in her analysis. In Duanmu's (2004) template (4), the free odd-numbered positions are strong. Whichever analysis is adopted, the positions of a line fall into two alternating classes: even-numbered positions are always strict, whereas only odd-numbered positions can be free (position 1 of pentasyllabic lines, and positions 1 and 3 of heptasyllabic lines), so that free positions are never adjacent. The analysis in Section 4 only refers to this parity-based alternation, and the constraints that encode generalisations in regulated verse (Section 4.4) do not consider the relative prominence between the two positions. The analysis is therefore compatible with both the iambic and the trochaic analyses.

(5) Arboreal analysis (adapted from Yip, 1980: 115–116)

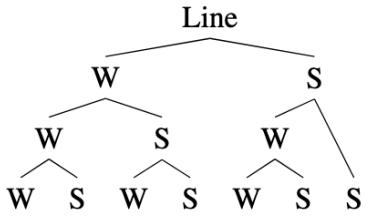
a. OOEEO and EEOOE lines:



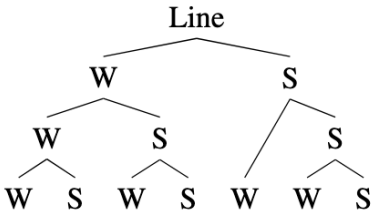
b. OOOEE and EEEOO lines:



c. OOEEOOE and EEOOEEO lines:



d. OOEEEOO and EEOOOEE lines:



(6) Grid analysis (adapted from Napoli, 1989: 28–29)

a. EEOOE lines:

```
          x  x
x  x  /   x  x  x
```

b. OOEEO lines:

```
x  x           x
x  x  /   x  x  x
```

c. EEEOO lines:

```
             x  x
x  x  /   x  x  x
```

d. OOOEE lines:

```
x  x      x
x  x  /   x  x  x
```

e. OOEEOOE lines:

```
x  x           x  x
x  x  x  x  /  x  x  x
```

f. EEOOEEO lines:

```
      x  x           x
x  x  x  x  /  x  x  x
```

g. OOEEEOO lines:

```
x  x              x  x
x  x  x  x  /  x  x  x
```

h. EEOOOEE lines:

```
      x  x     x
x  x  x  x  /  x  x  x
```

In brief, alternation is an important component of the tonal templates of regulated verse. On the one hand, a line may not contain too many syllables of the same category in a row, and the cadence of a canonical line contains both categories. On the other hand, the positions in each line are asymmetrical in their ability to tolerate tones from the opposite category: even-numbered positions are strict, and only odd-numbered positions can be free.

3.2 Non-alternation

Although some degree of tonal variation is necessary in regulated verse, the canonical templates also require that tone categories do not alternate too much. In the binary (iambic and trochaic) accentual and quantitative metres that have been studied most intensively, the regulated property alternates from position to position, so that a weak position is flanked by two strong ones and vice versa; even in ternary metres such as the dactylic hexameter, where two weak positions are adjacent, each strong position is flanked by weak ones. This is not true of regulated verse. The templates in (2) show that, except for the final position of each line, every position has at least one neighbouring position with a syllable from the same tone category. That is, if a position has an E syllable, at least one of the adjacent positions must also have an E syllable; the same applies to O syllables. In other words, a 'lone' tone category, i.e. an E flanked by two Os (OEO) or an O flanked by two Es (EOE), is not found in any non-final position of the canonical templates.

Free inversions can create such configurations, but only asymmetrically. Within the five-syllable core of a line, an inversion at position 1 of a type-A line creates a lone O (OOEEO → EOEEO), and such lines are fully regular, e.g. 烽火连三月 (平仄平平仄) in 杜甫's《春望》. In contrast, an inversion at position 1 of a type-B line, which would create a lone E (EEOOE → OEOOE), yields exactly the *guping* configuration that should be prohibited (Section 4). Within the core, lone E tones are found only in non-rhyming lines that traditional prosody regards as irregular (拗 *ao*) or as remedied, and never in accepted rhyming lines. The two syllables that heptasyllabic lines add before the core behave differently: an inversion at position

1 of a heptasyllabic line can leave a lone E at position 2, even in a rhyming line, as in 一行白鹭上青天 (仄平仄仄仄平平) in 杜甫's《绝句》or 此情可待成追忆 (仄平仄仄平平仄) in 李商隐's《锦瑟》. The non-alternation requirement therefore holds most strictly within the five-syllable core.

This pattern contrasts with accentual and quantitative metres, and even with other tonal metres. For example, the Vietnamese *lục bát* 'six-eight' metre (7), an indigenous Vietnamese verse form, also regulates the tone category allowed at each position of the line. It uses a binary classification whose terms are the Sino-Vietnamese counterparts of Even and Oblique: *bằng* 'flat' tones (the level *ngang* tone and the low-falling *huyền* tone) and *trắc* 'sharp' tones (the remaining four tones, *sắc*, *hỏi*, *ngã* and *nặng*, which are contour and/or glottalised tones). The *lục bát* metre dictates that even-numbered positions must use words from the specified category, whereas odd-numbered positions can use words from either category; these are marked as 'unregulated' in (7). Unlike in regulated verse, the unregulated positions show no preference for a tone category, and it is possible to find a flat tone surrounded by two sharp tones or vice versa, even in the second half of the line.

(7) Vietnamese *lục bát* metre (adapted from Kiparsky, 2020: 32)

× ♭ × ♯ × $♭_a$
× ♭ × ♯ × $♭_a$ ♯ $♭_b$
× ♭ × ♯ × $♭_b$
× ♭ × ♯ × $♭_b$ ♯ $♭_c$

(× = unregulated, ♭ = flat, ♯ = sharp, subscripts co-index required rhymes)

In short, although regulated verse requires tones to alternate within a line, such alternation cannot be too rapid: in the canonical templates, each non-final syllable has at least one neighbour from the same tone category. Hence, I propose that the metre of regulated verse requires alternation and non-alternation at the same time. To strengthen the argument, the next section examines how the two requirements interact when a line deviates from its template.

4. Case studies: 'Lone Even' and other remedies

In traditional scholarship on Chinese poetry, a line that deviates from its template at a position that does not allow inversion is called 'irregular' (拗 *ao*), and a compensating deviation elsewhere that renders such a line acceptable is called a 'remedy' (救 *jiu*). This section shows that the best-known irregularities and remedies demonstrate the interaction of alternation and non-alternation when analysed in Optimality Theory (OT; Prince & Smolensky, 1993). Section 4.1 discusses the definition and status of *guping*. Section 4.2 summarises the remedies described in traditional prosody. Section 4.3 briefly introduces OT, and Section 4.4 presents the constraints. Section 4.5 analyses *guping* and its two remedies, and Section 4.6 the

'three-inversion-four-remedy' pattern and the three-Even ending. Section 4.7 summarises the constraint ranking.

4.1 *Guping*: definitions and status

One can conclude from (2) that only the first position of pentasyllabic lines, and the first and third positions of heptasyllabic lines, allow inversions. However, this does not apply to all lines. The second line of Couplet-5O (type B) does not allow inversion in its first position, and the second line of Couplet-7E (also type B) does not allow inversion in its third position, which is unsurprising given that it is an extension of the former.

Traditional prosody attributes this restriction to the avoidance of *guping*. The rule was codified by Qing-dynasty prosodists. The《律诗定体》of 王士禛 states that, in the rhyming lines of pentasyllabic regulated verse in which the second and fourth positions are Even and Oblique, the first syllable must be Even, 'because the Even-Even [sequence] consists of only two syllables in a row and must not be made single' (第一字必用平，断不可杂以仄声，以平平止有二字相连，不可令单也; 丁福保, 1978: 113), and that the same holds for the third syllable of heptasyllabic lines. In modern scholarship, 王力 (1958, 1962) states the rule for the same line type: if the first syllable of 平平仄仄平 (or the third syllable of 仄仄平平仄仄平) is Oblique, the line contains only one Even syllable apart from the rhyme syllable, and this constitutes *guping*. His account has been widely followed.

However, both the definition and the status of *guping* have been debated. First, 启功 (2000: 26–27) defines *guping* configurationally, as an Even tone flanked by two Oblique tones (两仄夹一平), and treats it on a par with its mirror image *guze* 'Lone Oblique' (孤仄), an Oblique tone flanked by two Even tones; neither configuration yields a regular line. Second, the Heian-period Japanese manual for composing Chinese verse,《作文大体》, gives as a model of pentasyllabic regulated verse the pattern 平平他他平，平他他平平。平他平平他，他平他他平 (《作文大体》观智院本: 18), where 他 *ta* 'other' denotes the Oblique category. Its fourth line, 他平他他平 (OEOOE), is a *guping* line. It follows a line that begins with an Even tone, 平他平平他 (EOEEO), so that the two lines of the couplet are tonal opposites at every position. As 蔡瑜 (2019: 244–245) shows, this configuration follows from a rule of the early Tang prosodist 元兢, preserved in the《文镜秘府论》compiled by 空海. According to this rule, called 换头 'changing the head', the first syllables of successive lines should ideally alternate between the two categories; two Even first syllables are tolerated, but 'the first syllable of a line must not be rising, departing or entering when the first syllable of the next line is also rising, departing or entering, otherwise the tones are not harmonious' (又不得句头第一字是去上入，次句头用去上入，则声不调也; 元兢, in 蔡瑜, 2019: 232). If the second line of a couplet begins with an Oblique tone, the first line must therefore begin with an Even tone, and the resulting *guping* line is accepted. The prohibition of *guping* was thus not recognised in all traditions of regulated-verse composition. Third, the system of remedies (拗

救) in which the *guping* rule is embedded has been argued to be largely a construct of Qing scholars such as 赵执信 rather than a set of rules consciously observed by Tang poets (龚祖培, 2015), and its empirical basis has been re-examined against 15,290 regulated poems in the《全唐诗》(诸雨辰、胡韧奋, 2019).

These debates bear on the present analysis in two ways. First, I adopt the configurational definition of 启功: the canonical templates contain neither a lone E nor a lone O (Section 3.2), and I state both restrictions as general constraints (Section 4.4). The two configurations are not equally avoided, however. A lone O is freely created by inversion at position 1 of type-A lines (烽火连三月, 平仄平平仄), whereas a lone E is avoided in the five-syllable core of rhyming lines, the lines to which the definition of 王力 applies. I therefore add a more specific version of the restriction on a lone E that applies only to rhyming lines. Second, since the prohibition may not have been universal, I model the acceptance of *guping* in a couplet whose lines are tonal opposites in《作文大体》as another remedy besides the remedy within the line described by 王力. As shown in Section 4.5, a single grammar with variable ranking of violable constraints yields both of these remedies, as well as unremedied *guping*, which is also attested (Section 4.2).

4.2 Irregular lines and their remedies in traditional prosody

Using a word from the non-canonical category at a position that does not allow inversion is sometimes inevitable: the poet may, for instance, need a name or a particular word for which there is no alternative. Traditional prosody describes several ways of salvaging such lines (王力, 1958, 1962). The four configurations that are relevant here are summarised in (8), and attested examples are given in (9). Inverted positions are underlined.

(8) Irregularities and remedies (pentasyllabic forms; heptasyllabic forms in brackets)

a. *Guping* and its within-line remedy (孤平拗救, 本句自救).
   An inversion at position 1 of a type-B line (EEOOE → <u>O</u>EOOE) creates a lone E; it is remedied by an inversion at position 3 of the same line: <u>O</u>E<u>E</u>OE.
   [OOEEOOE → OO<u>O</u>EOOE → OO<u>O</u>E<u>E</u>OE]

b. *Guping* and the remedy in the other line of the couplet.
   The *guping* line <u>O</u>EOOE is accepted if the first syllable of the type-A line of the couplet is inverted as well (OOEEO → <u>E</u>OEEO), so that the two lines are tonal opposites at every position: <u>E</u>OEEO / <u>O</u>EOOE. This configuration is recorded in《作文大体》and follows from the rule of 换头 formulated by 元兢 (Section 4.1).
   [EE<u>E</u>OEEO / OO<u>O</u>EOOE]

c. Three-inversion-four-remedy (三拗四救).
   An inversion at position 3 of a type-C line (EEEOO → EE<u>O</u>OO) creates a three-Oblique ending; it is remedied by an inversion at position 4: EE<u>OE</u>O, which 王力 (1962)

describes as a special form of type C. The unremedied EEOOO is also attested but is dispreferred.

[OOEEEOO → OOEEOEO, 'five-inversion-six-remedy', 五拗六救]

d. Three-Even ending (三平调).

An inversion at position 3 of a type-D line (OOOEE → OOEEE) has no remedy; such lines are simply avoided. [position 5 of type D]

(9) Attested couplets

a. 跪进雕胡饭，月光明素盘。 仄仄平平仄，仄平平仄平 李白《宿五松山下荀媪家》
b. 何物随身去，六经与一琴。 平仄平平仄，仄平仄仄平 姚合《送进士田卓入华山》
c. 仍怜故乡水，万里送行舟。 平平仄平仄，仄仄仄平平 李白《渡荆门送别》
开轩面场圃，把酒话桑麻。 平平仄平仄，仄仄仄平平 孟浩然《过故人庄》
d. 时倚檐前树，远看原上村。 平仄平平仄，仄平平仄平 王维《辋川闲居》
e. 马眼冰凌影，竹根雪霰文。 仄仄平平仄，仄平仄仄平 李峤《绫》
晓出爬船寺，手擎紫栗条。 仄仄平平仄，仄平仄仄平 贾岛《寄乔侍郎》

Couplet (9a) illustrates the remedy within the line (8a). In (9b), position 1 of the rhyming line is O, so that the line contains a lone E (经). The line is not remedied within itself, but position 1 of the first line is E instead of O, so that the two lines are tonal opposites at every position, as in (8b). The two remedies thus differ in kind: the remedy within the line removes the lone E, whereas the remedy in the other line leaves it in place and restores the opposition between the two lines that the inversion destroyed. (9c) shows two examples of 'three-inversion-four-remedy'. (9d) and (9e) show two further configurations that the rules in (8) do not provide for. In (9d), the *guping* is remedied within the line, and the first syllable of the first line is E as well, so that the lines are opposite in their first two syllables. In (9e), the *guping* line is not remedied at all.

It is possible to list these rules and remedies as additions to the templates, but such a list has little explanatory power. It does not say why these positions resist inversion, why the remedies take the forms they do, why *guping* can be remedied in two different ways, why the three-Even ending has no remedy at all, or why it is the E category that cannot be 'lone'. In what follows, I argue that all of these properties follow from the interaction of constraints encoding alternation and non-alternation.

4.3 Basics of OT

OT is the theoretical framework initiated by Prince & Smolensky (1993) that relates the underlying representations (URs) of phonological forms to their surface forms (SFs) by means of universal violable constraints; differences among languages arise from differences in the ranking of these constraints. The constraints largely fall into two types: markedness constraints

penalise marked structures on the surface, and faithfulness constraints penalise mismatches between URs and SFs.

Potential surface realisations ('candidates') are evaluated on the basis of the violations they incur and the constraint hierarchy. For example, in a constraint hierarchy C1 ≫ C2 ≫ C3 (where ≫ means 'dominates'), satisfaction of C1 is categorically more important than satisfaction of C2, which is in turn more important than satisfaction of C3 (McCarthy, 2007). To evaluate the relative harmony of two candidates cand1 and cand2, one finds the most highly ranked constraint C that differentiates them; if cand1 violates C fewer times than cand2, cand1 is more harmonic than cand2. After all the candidates for a UR have been compared in this way, the most harmonic one is selected as the winner.

The tableaux[①] in (10) include two minimal examples. In (10a), the highest constraint that differentiates cand2 from cand1 is C1, because cand2 incurs one violation of C1 whereas cand1 does not. So, cand1 is more harmonic than cand2 and is the winner predicted by the mini grammar. Sometimes two candidates violate the same constraint a different number of times; then the one incurring fewer violations wins. Therefore, in (10b), cand2 loses to cand1 because it violates C2 twice whereas cand1 violates C2 only once.

(10) Examples of mini grammars in OT

a.

UR	C1	C2
☞ cand1		*
cand2	*!	

b.

UR	C1	C2
☞ cand1		*
cand2		**!

Although Prince & Smolensky (1993) propose that constraints are totally ranked, the available data usually cannot determine the total ranking, and some constraints may not be ranked with respect to each other by the time the winner is selected. In (11), cand3, which violates C1 once, loses to cand1, which violates only C3, because C1 outranks C3; similarly, cand2 loses to cand1 because C2 outranks C3. Since cand1 wins over both, the relative ranking between C1 and C2 is undetermined (indicated by the dashed line).

[①]The tableau notation follows Prince & Smolensky (1993): ☞ = winning candidate; * = violation; ! = fatal violation (i.e. a violation that eliminates the candidate from further comparison); a dashed line between two columns indicates that the two constraints are not ranked with respect to each other.

(11) Unranked constraints

UR	C1	C2	C3
☞ cand1			*
cand2		*!	
cand3	*!		

Two further notions are needed below. First, in some cases no candidate is acceptable. Following Prince & Smolensky (1993), I include among the candidates the null output ⊙, which leaves the input unrealised and violates only the constraint MPARSE. If every other candidate violates a constraint ranked above MPARSE, ⊙ wins; in metrics, this means that the line cannot be realised as intended and the poet must choose different words. Second, when two constraints are not ranked with respect to each other and their ranking matters for the outcome, the grammar can be understood as a set of total rankings, each of which selects its own winner. This captures variation between two outputs (Anttila, 1997).

4.4 Inputs, candidates and constraints

How does one determine the URs and SFs in metrics? If, in natural language, URs are understood as the canonical, abstract representations of morphophonemic forms and SFs as their possibly deviant realisations from the 'ideal' form, then, analogously, the requirements of a metre can be understood as the UR and the actual line patterns as SFs. In the case of regulated verse, the canonical templates in (2) are the inputs, and the possible variants of a template are the candidates. Since some of the regularities discussed below involve both lines of a couplet, I take the couplet as the unit of evaluation.

When nothing forces a deviation, the canonical template itself is optimal, since it violates none of the constraints below. The cases of interest arise when the poet chooses, at a position where the template does not allow it, a word whose tone category conflicts with the template. I treat this lexical choice as given: candidates that alter a lexically fixed tone category are not generated.[①] The question is then which of the candidates containing the forced inversion is optimal, i.e. what the best remedy is, or whether every such candidate is so bad that the line must be abandoned.

The constraints are given in (12). They are rooted in the generalisations of Section 3. Four of them, BIN-E, BIN-O, ODD-INVER and IDENT, are general constraints that state properties of the canonical templates. Most of the others are more specific versions of these constraints, restricted to rhyming lines or to particular positions.

[①]In standard OT practice, the fully faithful candidate is included in every tableau. Here the canonical template does not contain the lexically forced tone category and is therefore not considered a candidate. Using an undominated faithfulness constraint to lexical tone (IDENT-LEX) achieves the same effect. I have chosen to restrict the candidate set in this way and not to use a separate constraint, so IDENT-LEX is not listed in (12).

(12) Constraints and what they penalise

Constraint	Assign one violation for …
Markedness: non-alternation	
a. BIN-E	each non-final E that is flanked by two Os.
b. BIN-E(R)	each non-final E in the last five syllables of a rhyming line (a line whose template ends in E) that is flanked by two Os.
c. BIN-O	each non-final O that is flanked by two Es.
Markedness: alternation (variation)	
d. ALT-CAD	each line whose cadence (final three syllables) does not contain both tone categories (i.e. ends in EEE or OOO).
e. ALT-CAD(R)	each rhyming line whose cadence does not contain both tone categories (i.e. ends in EEE).
f. DUI (对)	each non-final position at which the two lines of a couplet carry the same tone category although the lines of the canonical couplet are opposite there.
Faithfulness	
g. IDENT	each inversion.
h. IDENT-CAD	each inversion in the cadence.
i. ODD-INVER	each inversion in an even-numbered position.
j. ODD-INVER(R)	each inversion in an even-numbered position of a rhyming line.
k. NO-INVER-H1	each inversion in an even-numbered position before the cadence (position 2 of pentasyllabic lines; positions 2 and 4 of heptasyllabic lines).
l. NO-INVER-FIN	each inversion in the final position of a line.
Other	
m. MPARSE	the null output ⊙.

BIN-E and BIN-O rephrase the generalisation of Section 3.2 that no non-final syllable of a canonical line is flanked by two syllables of the other category; they correspond to *guping* and its mirror image *guze* in the configurational sense of 启功 (Section 4.1). BIN-E(R) is the more specific version of BIN-E that applies to the five-syllable core of rhyming lines, i.e. lines whose template ends in E (types B and D), in line with the definition of *guping* by 王力 and with the requirement of 王士禛 that Even tones 'must not be made single'. BIN-O has no such specific version: a lone O is created freely by inversion at a free position, as in 烽火连三月 (EOEEO). This asymmetry will be discussed in Section 5.2.

ALT-CAD, ALT-CAD(R) and DUI encode alternation, understood as the avoidance of monotony. ALT-CAD states the generalisation of Section 3.1 that the cadence of a line contains both tone categories, and ALT-CAD(R) is its more specific version for rhyming lines, in which

the violating cadence is EEE, the three-Even ending.[①] DUI, named after the traditional term 对 *duì* 'opposition', encodes alternation between the two lines of a couplet. In the canonical couplets, the two lines carry opposite tone categories at every position: OOEEO / EEOOE and EEEOO / OOOEE, and likewise in heptasyllabic couplets. DUI requires this opposition to be preserved throughout the couplet. It assigns one violation for each position at which it is lost, so that it is fully satisfied only by a couplet whose lines are opposite at every such position, the configuration that traditional prosody calls 字字相对 (王力, 1958: 94).[②] Unlike the other markedness constraints, it can be satisfied by inversions in pairs: an inversion in one line is compensated by an inversion at the same position in the other line. DUI thus formalises the rule of 换头 formulated by 元兢 (Section 4.1).

IDENT penalises every inversion, and ODD-INVER every inversion in an even-numbered position. ODD-INVER encodes the parity-based alternation of strict and potentially free positions discussed in Section 3.1; since it refers to the parity of positions rather than to their prominence, it is neutral between the iambic and the trochaic analyses. ODD-INVER(R) protects all even-numbered positions of rhyming lines. NO-INVER-H1 protects the even-numbered positions before the cadence, which determine the line type and the relations of 粘 and 对 between lines. The only even-numbered position that can be inverted is therefore the penultimate position of a non-rhyming line, which is exactly where the three-inversion-four-remedy pattern is found. NO-INVER-FIN protects the final position of a line, and IDENT-CAD protects the cadence, where the metre is strictest (Section 3.1). No specific constraint protects the free positions before the cadence (position 1 of pentasyllabic lines; positions 1 and 3 of heptasyllabic lines). MPARSE penalises the null output.

IDENT, unlike the other three general constraints BIN-E, BIN-O and ODD-INVER, which are ranked at the bottom of the hierarchy (Section 4.7), is ranked below its specific version IDENT-CAD but not with respect to the other constraints in the lower stratum. It ensures that the canonical template is the only optimal candidate when nothing forces a deviation, and, where it dominates DUI, it prefers the candidate with fewer inversions to one whose lines are more opposite (Section 4.5).

[①]The all-Oblique cadence of non-rhyming lines (the three-Oblique ending, 三仄尾) violates only the general, violable ALT-CAD; unlike the three-Even ending, it is therefore not absolutely prohibited (Section 4.6).

[②]One could instead assign a single violation to every couplet that is not fully opposite. Such a version would not distinguish couplets that are opposite at more positions from couplets that are opposite at fewer, so that an inversion that restores opposition at one position but not at all of them would never be worthwhile. It could therefore not derive the remedy within the line with an E first syllable in the first line (Section 4.5), which is attested (9d) and which 元兢 considers the best form of 换头.

4.5 *Guping* and its two remedies[①]

Consider first a Couplet-5O in which the first syllable of the rhyming line is lexically O, as in 月光明素盘 (9a) and 六经与一琴 (9b). Four constraints in the lower stratum decide the outcome: BIN-E(R), IDENT-CAD, DUI and IDENT. They are only partially ranked: BIN-E(R) dominates DUI, and IDENT-CAD dominates IDENT, but no other ranking among them is fixed. Two choices therefore remain open. If BIN-E(R) dominates IDENT-CAD, the lone E is removed; otherwise it is tolerated. If DUI dominates IDENT, inversions that make the two lines (more) opposite are worthwhile; otherwise the candidate with fewer inversions wins. Every total ranking compatible with this partial ranking selects one of five candidates, (a)–(e), and tableaux (13)–(17) show one such ranking for each. In all of them, candidates (f) and (g), which avoid the lone E by inverting position 2 or the final position, and the null output (h) lose, since they violate constraints ranked above MPARSE or MPARSE itself. IDENT counts every inversion, including the forced one at position 1 of the second line, which all candidates except ⊙ share.

If BIN-E(R) ≫ IDENT-CAD ≫ IDENT ≫ DUI (13), the lone E is removed by an inversion at position 3, the only odd-numbered non-final position left; this is the traditional remedy within the line (本句自救). The first line is left unchanged: an E first syllable in the first line (13b) would make the lines opposite at position 1, but at the cost of an additional inversion. The winner (13a) is the familiar form of the remedy, as in 跪进雕胡饭，月光明素盘 (9a).

(13) Remedy within the line, BIN-E(R) ≫ IDENT-CAD ≫ IDENT ≫ DUI (input: OOEEO / EEOOE; line 2, position 1 lexically O)

	OOEEO / EEOOE	NO-INVER-FIN	ODD-INVER (R)	MPARSE	BIN-E (R)	IDENT-CAD	IDENT	DUI
a. ☞	OOEEO / OEEOE					*	**	**
b.	EOEEO / OEEOE					*	***!	*
c.	EOOEO / OEEOE					**!	****	
d.	EOEEO / OEOOE				*!		**	
e.	OOEEO / OEOOE				*!		*	*
f.	OOEEO / OOOOE		*!				**	**
g.	OOEEO / OEEOO	*!				**	***	**
h.	⊙			*!				

If DUI ranks above IDENT but below IDENT-CAD (14), the remedy within the line is accompanied by an inversion at position 1 of the first line, so that the lines in the same couplet

[①] The general constraints BIN-E, BIN-O and ODD-INVER are ranked below all the other violable constraints (Section 4.7). They therefore only decide between candidates that are equally good with respect to all the other constraints. Since they never do so in the cases discussed below, they are omitted from the tableaux in the rest of the paper.

are opposite in both their first and their second syllables (14b), as in 时倚檐前树，远看原上村 (9d). Since the remedy within the line does not restrict inversions in the first line, (14) is an acceptable extension to this remedy.[①] The lines remain non-opposite at position 3, since restoring the opposition there would require a further inversion in the cadence of the first line.

(14) Remedy within the line with an E first syllable in the first line, BIN-E(R) ≫ IDENT-CAD ≫ DUI ≫ IDENT (same input)

	OOEEO / EEOOE	NO-INVER-FIN	ODD-INVER (R)	MPARSE	BIN-E (R)	IDENT-CAD	DUI	IDENT
a.	OOEEO / OEEOE					*	**!	**
b. ☞	EOEEO / OEEOE					*	*	***
c.	EOOEO / OEEOE					**!		****
d.	EOEEO / OEOOE				*!			**
e.	OOEEO / OEOOE				*!		*	*
f.	OOEEO / OOOOE		*!				**	**
g.	OOEEO / OEEOO	*!				**	**	***
h.	⊙			*!				

If BIN-E(R) ≫ DUI ≫ IDENT-CAD ≫ IDENT (15), BIN-E(R) and DUI are both fully satisfied at the cost of two inversions at position 3 in the cadence and one at position 1. The result is a couplet whose lines are opposite in every syllable (字字相对).

(15) Lines opposite at every position, BIN-E(R) ≫ DUI ≫ IDENT-CAD ≫ IDENT (same input)

	OOEEO / EEOOE	NO-INVER-FIN	ODD-INVER (R)	MPARSE	BIN-E (R)	DUI	IDENT-CAD	IDENT
a.	OOEEO / OEEOE					*!*	*	**
b.	EOEEO / OEEOE					*!	*	***
c. ☞	EOOEO / OEEOE						**	****
d.	EOEEO / OEOOE				*!			**
e.	OOEEO / OEOOE				*!	*		*
f.	OOEEO / OOOOE		*!			**		**
g.	OOEEO / OEEOO	*!				**	**	***
h.	⊙			*!				

[①] In fact, making both the first and second syllables of the two adjacent lines opposite is what 元兢 calls the 'double change of the head' (双换头), which he considers best (是最善也), whereas having opposite tone categories in the second syllable only 'is not as good as the double change' (不及双换; 元兢, in 蔡瑜, 2019: 232).

If IDENT-CAD dominates BIN-E(R) (16), the cadence is kept intact and the lone E surfaces as it is. If DUI dominates IDENT, the opposition between the lines can still be restored by an inversion at position 1 of the first line, which makes the two lines opposite at every position (16d). This is the remedy in the other line recorded in《作文大体》and required by the rule of 换头. The additional inversion does create a lone O in the first line (EOEEO), but this violates only the general BIN-O, which is ranked at the bottom of the hierarchy.

(16) Remedy in the other line, IDENT-CAD ≫ BIN-E(R) ≫ DUI ≫ IDENT (same input)

	OOEEO / EEOOE	NO-INVER-FIN	ODD-INVER (R)	MPARSE	IDENT-CAD	BIN-E (R)	DUI	IDENT
a.	OOEEO / OEEOE				*!		**	**
b.	EOEEO / OEEOE				*!		*	***
c.	EOOEO / OEEOE				*!*			****
d. ☞	EOEEO / OEOOE					*		**
e.	OOEEO / OEOOE					*	*!	*
f.	OOEEO / OOOOE		*!				**	**
g.	OOEEO / OEEOO	*!			**		**	***
h.	⊙			*!				

Finally, if IDENT-CAD ≫ BIN-E(R) ≫ IDENT ≫ DUI (17), neither the lone E nor the lost opposition can be repaired, and the *guping* line surfaces unremedied (17e), as in 马眼冰凌影，竹根雪霰文 (9e).

(17) Unremedied *guping*, IDENT-CAD ≫ BIN-E(R) ≫ IDENT ≫ DUI (same input)

	OOEEO / EEOOE	NO-INVER-FIN	ODD-INVER (R)	MPARSE	IDENT-CAD	BIN-E (R)	IDENT	DUI
a.	OOEEO / OEEOE				*!		**	**
b.	EOEEO / OEEOE				*!		***	*
c.	EOOEO / OEEOE				*!*		****	
d.	EOEEO / OEOOE					*	**!	
e. ☞	OOEEO / OEOOE					*	*	*
f.	OOEEO / OOOOE		*!				**	**
g.	OOEEO / OEEOO	*!			**		***	**
h.	⊙			*!				

The configurations of *guping* attested in (9) are thus outputs of a single grammar. They differ in which requirements are satisfied: non-alternation within the line (BIN-E(R)), alternation between the lines (DUI), or faithfulness to the template (IDENT-CAD, IDENT). The pairs (13a) and (14b), and (16d) and (17e), differ only in the ranking of DUI and IDENT, i.e. in whether an inversion at the free first position of the first line is worth making for the sake of opposition. This also answers the question of why a second deviation can make a *guping* line acceptable: the inversion in the first line in (16d) does not repair the lone E but restores the opposition between the lines that the forced inversion destroyed. A lone E, however, is never created for the sake of opposition, because BIN-E(R) dominates DUI. In 烽火连三月，家书抵万金 (EOEEO / EEOOE), for instance, the free inversion at position 1 of the first line is not matched by an inversion at position 1 of the second line (*EOEEO / OEOOE), although this would make the lines fully opposite. When position 3 of the rhyming line is also lexically O (i.e. the remedy within the line is unavailable), only the remedy in the other line and the unremedied couplet remain. The heptasyllabic case is parallel, with the forced inversion at position 3, the remedy within the line at position 5 and the inversion that restores the opposition at position 3 of the first line.

4.6 Three-inversion-four-remedy and the three-Even ending

Consider next a Couplet-5E in which position 3 of the non-rhyming type-C line is lexically O, as in 仍怜故乡水 and 开轩面场圃 (9c). Tableaux (18a) and (18b) show the evaluation under two rankings of ALT-CAD, IDENT-CAD and DUI.

(18) Three-inversion-four-remedy in Couplet-5E (input: EEEOO / OOOEE; line 1, position 3 lexically O)

a. ALT-CAD ≫ IDENT-CAD ≫ DUI ≫ IDENT

	EEEOO / OOOEE	ALT-CAD(R)	MPARSE	ALT-CAD	IDENT-CAD	DUI	IDENT
i. ☞	EEOEO / OOOEE				**	**	**
ii.	EEOOO / OOOEE			*!	*	*	*
iii.	EEOOO / OOEEE	*!		**	**		**
iv.	⊙		*!				

b. IDENT-CAD ≫ ALT-CAD ≫ DUI ≫ IDENT

	EEEOO / OOOEE	ALT-CAD(R)	MPARSE	IDENT-CAD	ALT-CAD	DUI	IDENT
i.	EEOEO / OOOEE			**!		**	**
ii. ☞	EEOOO / OOOEE			*	*	*	*
iii.	EEOOO / OOEEE	*!		**	**		**
iv.	⊙		*!				

The winner in (18a), EEOEO (平平仄平仄), deviates more from the template than the unremedied EEOOO, and it violates more constraints: it contains a lone E at position 4 and a lone O at position 3, and it inverts an even-numbered position. Most of these violations concern only the general constraints BIN-E, BIN-O and ODD-INVER, which are ranked at the bottom of the hierarchy, and not BIN-E(R), since type-C lines do not rhyme. The inversion at position 4 also makes the lines non-opposite at that position, in violation of DUI, and adds an inversion. The unremedied EEOOO, in turn, is not violation-free: its cadence contains no E, in violation of ALT-CAD. If ALT-CAD dominates IDENT-CAD and DUI (18a), the line restores alternation in its cadence by an inversion at position 4, even though this adds violations of IDENT-CAD and DUI. Alternation cannot be restored in the other line of the couplet (candidate iii in (18)): ALT-CAD is evaluated line by line, and inverting position 3 of the type-D line would in any case produce a three-Even ending, which is fatal. If IDENT-CAD dominates ALT-CAD, as in (18b), or DUI does, the unremedied three-Oblique ending wins. Both outcomes are attested: the three-inversion-four-remedy pattern is a widespread variant of type C, while the three-Oblique ending is dispreferred but not prohibited (Section 3.1). The heptasyllabic counterpart, 'five-inversion-six-remedy' (OOEEEOO → OOEEOEO), is derived in the same way.

The three-Even ending illustrates a third possibility. Suppose that position 3 of a type-D (rhyming) line is lexically E (19).

(19) Three-Even ending in Couplet-5E (input line: OOOEE; position 3 lexically E)

	OOOEE	NO-INVER-FIN	NO-INVER-H1	ALT-CAD (R)	ODD-INVER(R)	MPARSE	BIN-E(R)
a.	OOEEE			*!			
b.	OOEOE				*!		*
c.	OEEOE		*!		**		
d.	OOEEO	*!					
e. ☞	⊙					*	

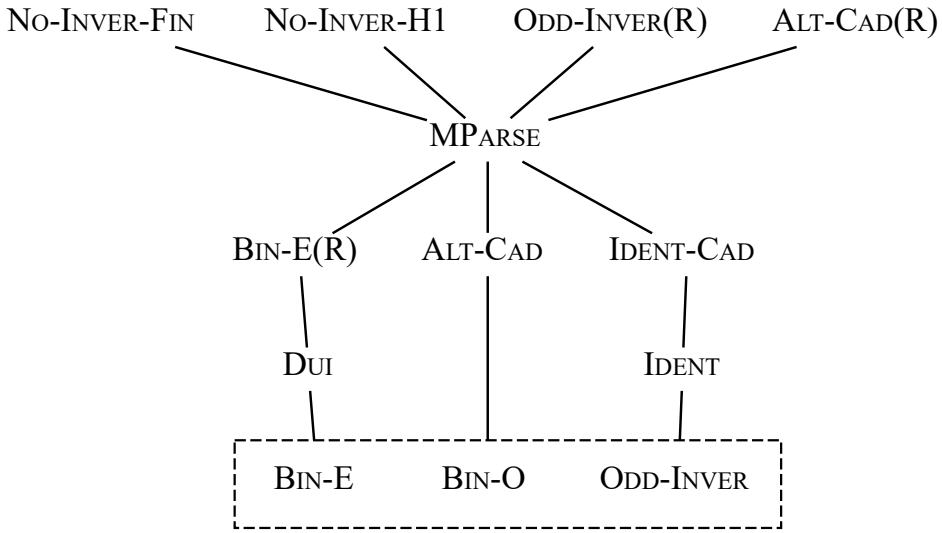
The faithful realisation OOEEE (19a) violates ALT-CAD(R). Inverting position 4 (19b) would restore alternation in the cadence, but it would invert an even-numbered position of a

rhyming line (and create a lone E); inverting positions 2 and 4 (19c) would turn the line into a different line type; and inverting the final position (19d) would destroy the rhyme. Since all of these constraints rank above MPARSE, the null output wins: the line has no remedy and must be avoided, exactly as traditional prosody prescribes.

4.7 Summary

The ranking established in Sections 4.5 and 4.6 is shown as a Hasse diagram in (20). A line connecting two constraints indicates that the higher one dominates the lower one, and constraints that are not connected by a descending path are not ranked with respect to each other. The three general constraints in the dashed box are dominated by every constraint above them and are not ranked with respect to each other. The grammar is the set of total rankings compatible with (20).

(20) Ranking of the constraints (Hasse diagram)



The constraints above MPARSE are never violated: they rule out candidates such as (19a–d), and when every repair is blocked, the null output wins. Below MPARSE, BIN-E(R) dominates DUI, so that a lone E is never introduced for the sake of opposition, and IDENT-CAD dominates IDENT, as the metre is strictest in the cadence. DUI dominates the general constraint BIN-O; otherwise the remedy in the other line, which adds a lone O, could not win. The variation in (13)–(17) follows from the absence of a ranking between BIN-E(R) and IDENT-CAD and between DUI and IDENT (and IDENT-CAD), and the variation in (18) from the absence of a ranking between ALT-CAD on the one hand and IDENT-CAD and DUI on the other.

The analysis shows that the remedies of regulated verse result from a balance between alternation and non-alternation. Alternation in the cadence of rhyming lines (ALT-CAD(R)) is never sacrificed, whereas non-alternation (BIN-E(R)), alternation in the cadence of non-rhyming lines (ALT-CAD) and alternation between the lines of a couplet (DUI) are violable and

only partially ranked. The ranking of BIN-E(R) and IDENT-CAD decides whether *guping* is removed, in favour of non-alternation, or tolerated, and the ranking of DUI and IDENT decides whether the lines are made (more) opposite, in favour of alternation, or left as they are. Neither type of requirement is uniformly dominant, and regulated verse is the result of a balance between alternation and non-alternation.

5. Discussion

The previous sections have shown, through the patterns of the canonical templates and the remedies for irregular lines, that regulated verse requires the tones in each line both to alternate and not to alternate. In this section, I discuss how regulated verse contributes to the study of metres and to our knowledge of natural language phonology, and I address two questions about the status of the analysis: whether it describes or explains the patterns, and whether the *guping* restriction is a convention or an intuition.

5.1 Metre typology

First, the metre of regulated verse enriches the typology of metres. Most of the metres analysed in detail are syllabic, accentual or quantitative metres of Indo-European languages. Accentual and quantitative metres require the property they regulate, stress or weight, to vary: a line consisting entirely of unstressed or stressed syllables, or entirely of light or heavy syllables, is rarely if ever metrical. The tonal metre of regulated verse resembles accentual and quantitative metres in that the tone categories of the syllables in a line also need to vary to some extent. As shown in Section 3.1, the maximum number of syllables of the same category in a row in the canonical templates is three, and their cadences contain both categories. Regulated verse is thus consistent with the view that metres regulate variation (Kiparsky, 2020). Regulated verse also behaves like accentual and quantitative metres in that its positions are inherently asymmetrical: the freedom to use the non-canonical tone category differs systematically between odd- and even-numbered positions, an asymmetry that has been related to the prominence alternation of accentual and quantitative metres, whichever positions are taken to be strong (Section 3.1).

On the other hand, the metre of regulated verse differs significantly from previously studied metres. In accentual and quantitative metres, the regulated property alternates between adjacent positions: in iambic and trochaic metres, an unstressed or light syllable is typically flanked by two stressed or heavy ones, and vice versa, and even in ternary metres each strong position is flanked by weak ones. The tonal metre of regulated verse does the opposite: in its canonical templates, no non-final syllable is flanked by two syllables of the other category. Although regulated verse also requires tones to alternate, it prefers them to change gradually rather than abruptly.

The enrichment of the typology is not restricted to metres for non-tonal languages. Other tonal metres differ from regulated verse with respect to the categories of adjacent positions. As shown in Section 3.2, Vietnamese *lục bát*, which also regulates the tone categories of even-numbered positions more strictly, does not require syllables to have a neighbour from the same category. In other words, non-alternation is neither an absolute requirement nor an inherent property of tonal metres. Regulated verse therefore provides a novel angle for phonologists studying metres, namely that non-alternation must be balanced with alternation in tonal metres.

5.2 Natural language and metre

Second, the metre of regulated verse reflects several features of natural languages with tones. The fact that the canonical templates never contain more than three consecutive syllables from the same tone category resembles restrictions on the number of tone-bearing units (TBUs) that a tone may be associated with in many tonal languages. For example, many Bantu languages exhibit a bounded tone-spreading process called 'tone tripling', in which a high tone[①] spreads to up to two following TBUs within a word, but no further, as exemplified in (21) with Shona. This process is also found in Kanye Tswana (Creissels, 1998), Venda, Kalanga, Tsonga and Copperbelt Bemba (Odden & Marlo, 2019), and Dembwa Taita I (Odden, 2001). In Prinmi, a Tibeto-Burman language spoken in China, no more than two H-toned TBUs and no more than three L-toned TBUs are allowed in a word (Ding, 2006; Hyman, 2009; Jardine, 2020). (22) provides an example containing three consecutive L-toned TBUs and one containing two consecutive H-toned TBUs, with their tonal patterns given on the right.

(21) Shona H tone tripling (Odden, 2015)

kutóra	'to take'
kutórésá	'to make take'
kutóréséra	'to make take for'
kutóréserana	'to make take for each other'

(22) Prinmi tone spreading restrictions (Ding, 2006: 14)

bíbˈobˈoge	'as for roasted flour'	HLLL
tõpúmɜ́łe	'donkey tail'	LHHL

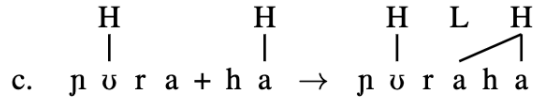
Another feature of the metre of regulated verse that resembles tone in natural languages is the avoidance of rapid excursions in pitch (McPherson, 2016). In regulated verse, an Even tone flanked by two Oblique tones, or an Oblique tone flanked by two Even tones, is absent from the canonical templates, suggesting that the change of tone categories cannot be too abrupt. Similarly, many tonal languages prohibit a tone surrounded by two tones of a different type, as captured by constraints such as *HLH (Cahill, 2007) or *TROUGH (Yip, 2002: 137). For

[①] In (21) and (22), high tones are indicated with an acute accent; all unmarked TBUs are low-toned.

example, in Kɔnni (Gur, Ghana), when a sequence HLH would be expected to arise from the combination of morphemes, the second H spreads leftwards, delinking the L and causing downstep (23). Other languages that disprefer HLH sequences include Deg, Ga, Esaaka Makhuwa and Zulu in Africa (Odden, 2000), Mianmin and Kairi in Papua New Guinea (Cahill, 2011), and Iñapari (Arawakan, Peru) and Mamaindê (Nambikwaran, Brazil) in South America (Parker, 1999; Eberhard, 2007).

(23) Kɔnni disprefers HLH sequences (Cahill, 2007)

a. ɲórà ‘chests’
b. ɲó$^{!}$ráhá ‘the chests’



Even Chinese has a similar phenomenon. In (Beijing) Mandarin, a dip in pitch between two high tone targets tends to be eliminated (Chen, 2000). T2 sandhi stipulates that a non-final rising tone [35] becomes a high tone [55] when preceded by a rising [35] or high [55] tone (24), since each of the sequences [55.35] and [35.35] contains a dip in pitch between two high targets. Similarly, the negator *bù* (不), whose citation tone is a falling [51], changes to a rising [35] when followed by another falling [51] syllable (25a), which eliminates the dip in [51.51]. This change is lexically restricted, however: it is part of the so-called *yi-bu-qi-ba* rule, which affects only a handful of morphemes, and an otherwise identical sequence such as *bùduì* ‘troops’ (部队) does not change (25b). It thus shows the same preference, but only in a lexically restricted way.

(24) T2 sandhi (Chen, 2000: 21–22)

a. *tiānwéntái* (天文台) ‘observatory’

55.35.35	base form
55.55.35	SF (with T2 sandhi)

b. *rénmínbì* (人民币) ‘Renminbi (Chinese currency)’

35.35.51	base form
35.55.51	SF (with T2 sandhi)

(25) Minimal pair showing the lexical restriction of the *yi-bu-qi-ba* rule (Chang, 1992: 166f; Chen, 2000: 22)

a. *búduì* (不对) ‘not correct’

51.51	base form
35.51	SF (with *yi-bu-qi-ba* rule)

b. *bùduì* (部队) ‘troops’

51.51	base form
51.51	SF (no change)

An interesting asymmetry between H and L tones exists within single languages as well as cross-linguistically: HLH is commonly avoided, whereas LHL occurs abundantly. Cahill (2007) attributes this asymmetry to the markedness of H tones, given that systems in which H is the marked tone are more frequent than those in which L is marked (Maddieson, 1978). The

asymmetry seems to have a parallel in regulated verse. There is a rule that prohibits a lone Even tone in a line, namely the rule of *guping*. It is implemented much more strictly than the avoidance of lone Oblique tones, which are freely created by free inversions (烽火连三月).

One might ask why this parallel presupposes that the Even category corresponds to L and the Oblique category to H in the natural-language pattern. At face value, Even tones might resemble L tones in the sense that they are both lower in pitch. One of the earliest phonetic descriptions of the Middle Chinese tones, recorded by the Japanese monk Annen (安然) in his *Shittanzō* (880 CE; 悉昙藏), describes the Even tone as 'level and low' (平声直低) and the rising tone as 'straight and high' (上声直昂) (丁邦新, 1975: 11), which, at least for the variety described, places the Even tone lower in pitch than the rising tone. But this is not a conclusive argument because the phonetic values of the Middle Chinese tones remain uncertain (Mei, 1970).

Instead, the parallel between natural language and regulated verse concerns markedness. Cahill's (2007) account of the HLH/LHL asymmetry rests on the markedness of H: what is avoided is a trough of the unmarked tone between two marked tones, whereas a peak of the marked tone between two unmarked ones is tolerated. The parallel thus requires that Even be the unmarked member of the Even/Oblique opposition, rather than that Even be phonetically low. There are two reasons to regard Even as the unmarked member. First, on the basis of the earliest descriptions, 丁邦新 (1975) argues that the opposition is fundamentally one between the Even tone, a level tone of normal length, and non-level (i.e. contour) tones: a high rising tone (上), a mid falling tone (去) and a short checked tone (入) that occurred only in syllables closed by a stop. Cross-linguistically, contour tones are marked relative to level tones: they are less frequent in tone inventories and tend to be restricted to prosodically privileged positions; checked syllables, for their part, typically allow fewer tonal contrasts (Maddieson, 1978; Yip, 2002; Zhang, 2002). Second, Even is the default category of the metre: rhymes in regulated verse are almost always Even, and the Japanese tradition represented by《作文大体》refers to the Oblique category simply as 他 *ta* 'other', defining it negatively with respect to Even. The default status of Even could arguably be the result of its unmarkedness.

In brief, several features of the metre of regulated verse have clear parallels in natural languages with tones, suggesting that metres are not random choices by poets but are heavily informed by natural language phonology. More importantly, these features are not confined to Chinese but are widely observed in tonal languages around the world. The metrical characteristics of regulated verse are therefore arguably a reflection of cross-linguistic preferences in tonal phonology.

5.3 Description or explanation?

Does the OT analysis describe the patterns of regulated verse, or explain them? It describes them, in that a ranking of constraints generates the attested lines and excludes the unattested ones, but it goes beyond a restatement of the traditional rules in three respects. First, it is more unified. The location of each remedy follows from constraints that do not refer to particular remedies: the remedy within the line is at position 3 because that is the only free position left; the remedy in the other line is at position 1 because that is where the opposition between the lines was lost; the three-inversion-four-remedy pattern stays within the line because alternation in the cadence is evaluated line by line; and the three-Even ending has no remedy because every repair violates an inviolable constraint. Second, the constraints are not specific to regulated verse: constraints against tonal troughs (Section 5.2), the preference for variation within metrical units, parallelism between the lines of a couplet (对仗) and positional faithfulness are all found elsewhere. Third, the analysis makes testable predictions that the rules do not make. An inversion in the first line should accompany a *guping* line only where it restores the opposition to the second line, i.e. at position 1, or at position 3 if the second line is remedied; the first syllable of the first line should therefore be Even more often before a rhyming line with an Oblique first syllable than before one with an Even first syllable, whether or not the *guping* is remedied; and no tradition of the metre should penalise a lone Oblique tone while tolerating a lone Even tone.

The analysis does not, however, explain why the constraints are ranked as they are. The restriction of the strictest constraints to rhyming lines, for instance, is stated rather than derived, and an explanation of the asymmetry between Even and Oblique must come from outside the grammar of the metre, for example from the markedness considerations of Section 5.2. The analysis is thus explanatory to the extent that it reduces the traditional rules to a small set of independently motivated constraints, and descriptive with respect to the ranking itself.

5.4 Convention or intuition? The status of *guping*

Is the *guping* restriction an artificial convention, comparable to those of the eight-legged essay (八股文), or is it grounded in native intuition? Its absence from Old Style and folk verse does not show that it is arbitrary: metrical rules are specific to verse forms, and Old Style verse does not use the templates of regulated verse at all (the three-Even ending, prohibited in regulated verse, is common there). All metres are learned conventions, but the space of possible conventions is constrained by linguistic structure (Fabb & Halle, 2008; Kiparsky, 2020), and the content of the restriction, a ban on an isolated Even tone in the strictest line of the couplet together with an asymmetry between Even and Oblique, mirrors restrictions in the tonal phonology of unrelated languages (Section 5.2). In other words, these artificial requirements do not come out of nowhere, and what the poets consider acceptable is grounded in and constrained by the phonology of their language.

The analysis also clarifies the debate on its status (Section 4.1): *guping* is not an absolute prohibition but the effect of a violable constraint whose ranking with respect to IDENT-CAD varies. Under one ranking it is remedied within the line, the norm codified by Qing prosodists and by 王力; under the other it is tolerated, either in a couplet whose lines are opposites, as in《作文大体》, or without any compensation (9e). Corpus counts such as those of 诸雨辰、胡韧奋 (2019) can show how often each ranking was chosen, and whether the lines that precede *guping* lines begin with an Even tone more often than expected (Section 5.3). That the earlier system accepts *guping* while the later one prohibits it is also consistent with the view that the prohibition is a convention that hardened over time, whereas the preference for opposition between the lines of a couplet is older and more general.

The restriction could also be tested experimentally. Since the entering tone has merged with the other tones in Modern Standard Chinese, the participants would be readers trained in regulated verse, who know the Middle Chinese tone categories, and untrained speakers of a variety that preserves the entering tone, such as Cantonese or Southern Min. In an acceptability-judgement task, they could rate rhyming pentasyllabic lines with *guping*, with each of its remedies, and with no irregularity. If the restriction is a learned convention, only the trained group should disprefer OEOOE; if it reflects a general dispreference for tonal troughs, the untrained group should disprefer it too, at least in speeded tasks that minimise explicit metrical knowledge. Corpus-based models of gradient well-formedness (Hayes, et al., 2012) offer a complementary approach. I leave this question for future research.

6. Conclusion

This paper has proposed a principle of balancing non-alternation and alternation in tonal metres, with evidence from the metre of Middle Chinese regulated verse. The canonical templates require tone categories to alternate: no more than three syllables of the same category occur in a row, the cadence of every line contains both categories, and strict and potentially free positions alternate by parity. At the same time, they require tone categories not to alternate too abruptly, since no non-final syllable of a canonical line is flanked by two syllables of the other category. In an OT analysis of the traditional rules for irregular lines, a single partially ranked grammar derives *guping*, both of its remedies and their attested variants, the three-inversion-four-remedy pattern and the unremediable three-Even ending. The remedy within the line satisfies non-alternation (BIN-E(R)), whereas the remedy in the other line, recorded in《作文大体》, satisfies alternation between the lines (DUI); the two are outputs of different rankings of the same constraints. Because the two kinds of constraints are interleaved and partly unranked, neither is uniformly dominant: they are balanced. The analysis relies only on the linear sequence of tone categories, the cadence, the parity of positions and the opposition between the lines of a couplet, and is therefore independent of the debate on foot headedness.

These characteristics of regulated verse resemble the phonology of tonal languages in general. The ‘up-to-three’ restriction on runs of the same tone category is similar to bounded tone spreading, such as tone tripling, and the avoidance of a lone Even tone, but not of a lone Oblique tone, patterns with the avoidance of tonal troughs in tonal languages, including Mandarin, provided that the Even category is the unmarked member of the opposition. Regulated verse is thus another case showing that metres are not merely a matter of poets’ personal preferences, but a balance between the natural tendency of tones and the rhythmic or prosodic alternation required in poetry.

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声调格律系统中“非交替”与“交替”的平衡原则

摘要：诗歌格律制约诗篇及诗行的基本节律格局，并通常要求其所规约的音系属性（如重音、音节重量或声调）在诗行内部交替分布，以避免单调。本文以中古汉语近体诗的平仄格律为例，提出格律不仅可以要求交替，也可以要求非交替，且二者须相互平衡。近体诗的标准律式要求句内平仄交替，但交替只在受限范围内实现：律式中任何非句末音节都不处于两个异类声调之间。在标准律式之外，本文在优选论框架下说明，传统诗律学针对拗句所规定的“救”法源于表征交替与非交替的制约条件之间的相互作用。孤平的两种救法是同一个部分排序语法在不同等级排列下的输出：本句自救满足禁止孤平的“非交替”制约条件；而在同联另一句中补救（出句首字改用平声，使上下两句平仄字字相对）则满足要求一联两句平仄相对（对）的“交替”制约条件，后者见于日本平安时代的《作文大体》。“三拗四救”则源于句末三字的“交替”要求优先于忠实性要求的等级排列。最后，本文指出，这一平衡性质丰富了诗歌格律的类型学；而只要将平声理解为平仄对立中无标记的（平调）一方，近体诗格律就与自然语言声调音系中广泛存在的限制相平行。

关键词：格律　声调语言　孤平　汉语诗歌　近体诗

[中文作者姓名、作者单位（具体到学院）及英文作者信息、作者邮箱：匿名评审稿中略去]